

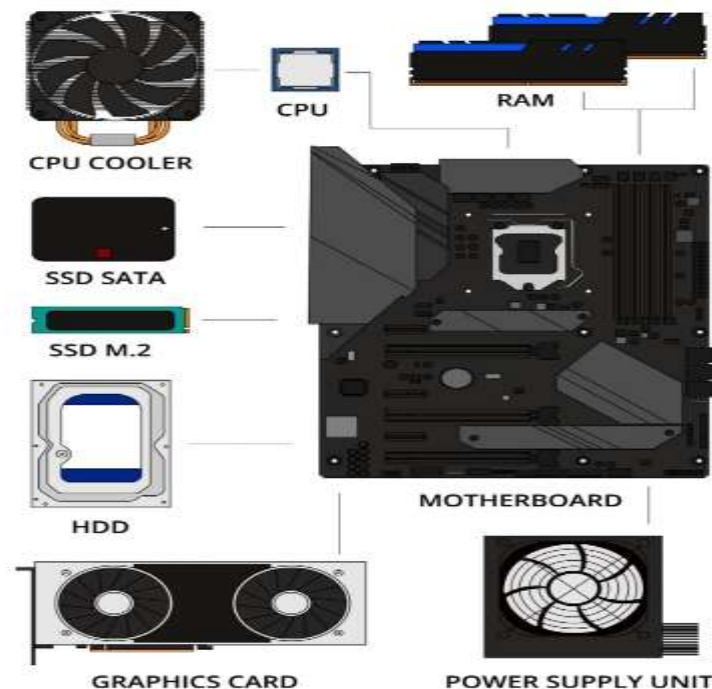
Question:1

Draw a labeled diagram of a desktop PC showing the motherboard, CPU, RAM, GPU, storage (HDD/SSD), power supply, and cooling system. For each component, write its main function next to it.

Answer: 1

Main Functions of PC Components

- **Motherboard:** It connects all the components together and allows them to communicate.
- **CPU (Central Processing Unit):** It acts as the "brain" of the computer and executes all instructions.
- **RAM (Random Access Memory):** It holds temporary data for currently running apps so the CPU can access it quickly.
- **GPU (Graphics Card):** It handles all visual processing and displays images/videos on the monitor.
- **Storage (HDD/SSD):** It permanently saves all your files, games, and the operating system.
- **Power Supply (PSU):** It converts wall electricity into safe power for all PC parts.
- **Cooling System:** It removes heat from the components to prevent the PC from overheating.



Question: 2

Create a table listing each major PC component (motherboard, CPU, RAM, GPU, storage, power supply, cooling system), its typical location inside the case, and one example of how it connects or interacts with another component.

Hint: For example, "CPU sits in the motherboard socket and communicates with RAM via the memory slots."

Answer: 2

Component	Location inside the Case	Connection / Interaction
Motherboard	Fixed inside the case	It connects all the main components.
CPU	In the CPU socket on the motherboard	It works with RAM to process data.
RAM	In the RAM slots on the motherboard	It gives data to the CPU quickly.
GPU	In a PCIe slot on the motherboard	It processes graphics and sends them to the monitor.
Storage (HDD/SSD)	In a drive bay or M.2 slot	It stores files and the operating system.
Power Supply (PSU)	Usually at the bottom of the case	It gives power to all the components.
Cooling System	Around the CPU and different places in the case	It removes heat and keeps the PC cool.

Question: 3

Take a clear photo or screenshot of the inside of any desktop PC (your own, a friend's, or an image from the web) and use any image editor to digitally label at least 6 major components as discussed in class.

Answer: 3



Question: 4

Write a short explanation (3-4 sentences) describing the data flow when you open a game on your PC, mentioning the roles of CPU, RAM, storage, GPU, and motherboard.

***Hint:** Start from storage loading data, then CPU processing, RAM usage, and GPU rendering.*

Answer: 4

When you open a game , the **motherboard** acts as the main highway connecting all parts so they can share data . First, the game files are loaded from your **storage** (SDD/HDD) directly into the **RAM** for fast , temporary access. Next , the **CPU** acts as the brain to process the game's logic, rules, and your keyboard inputs. Finally, the **GPU** takes this processed data and renders it into the actual graphics and pictures you see on your screen.